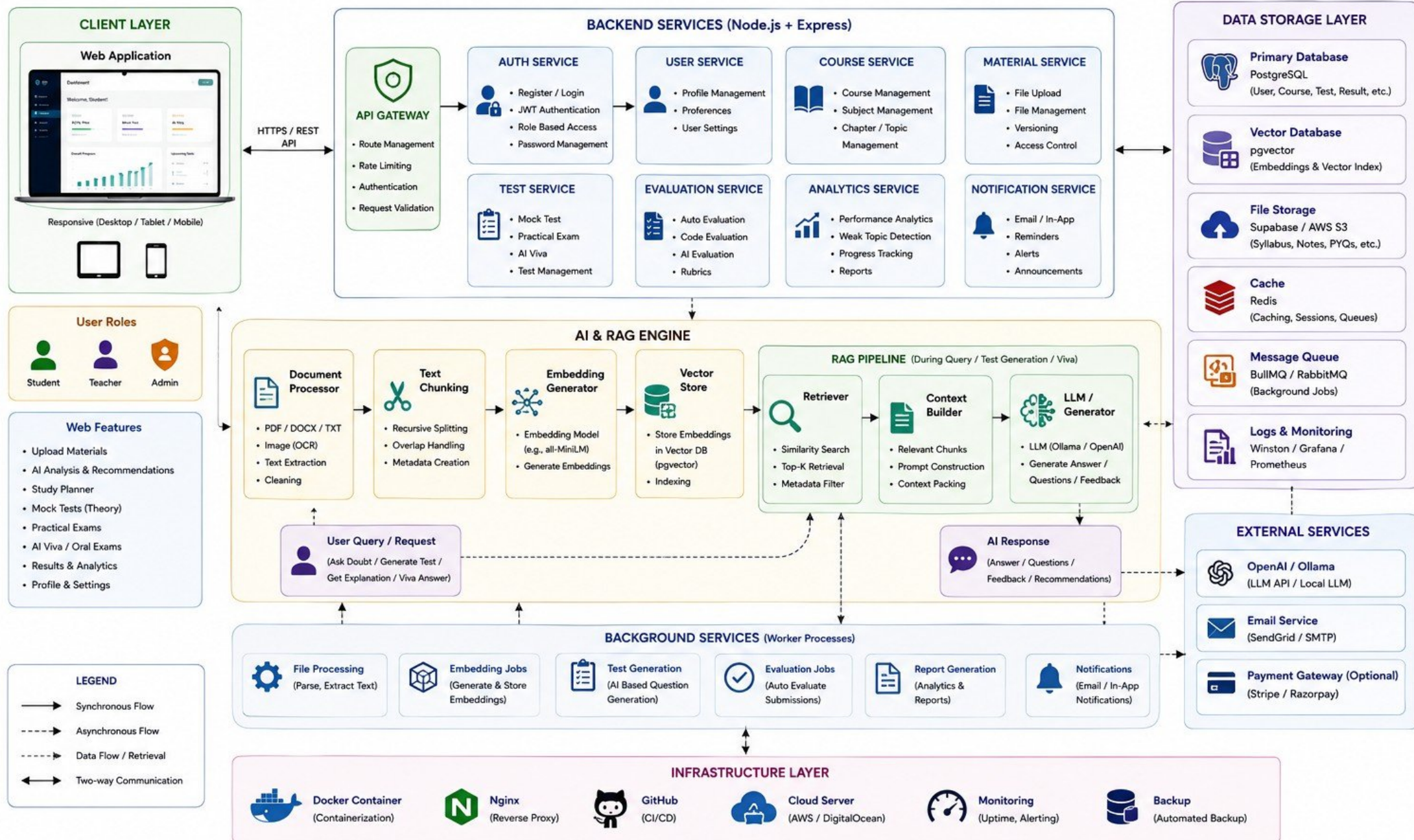
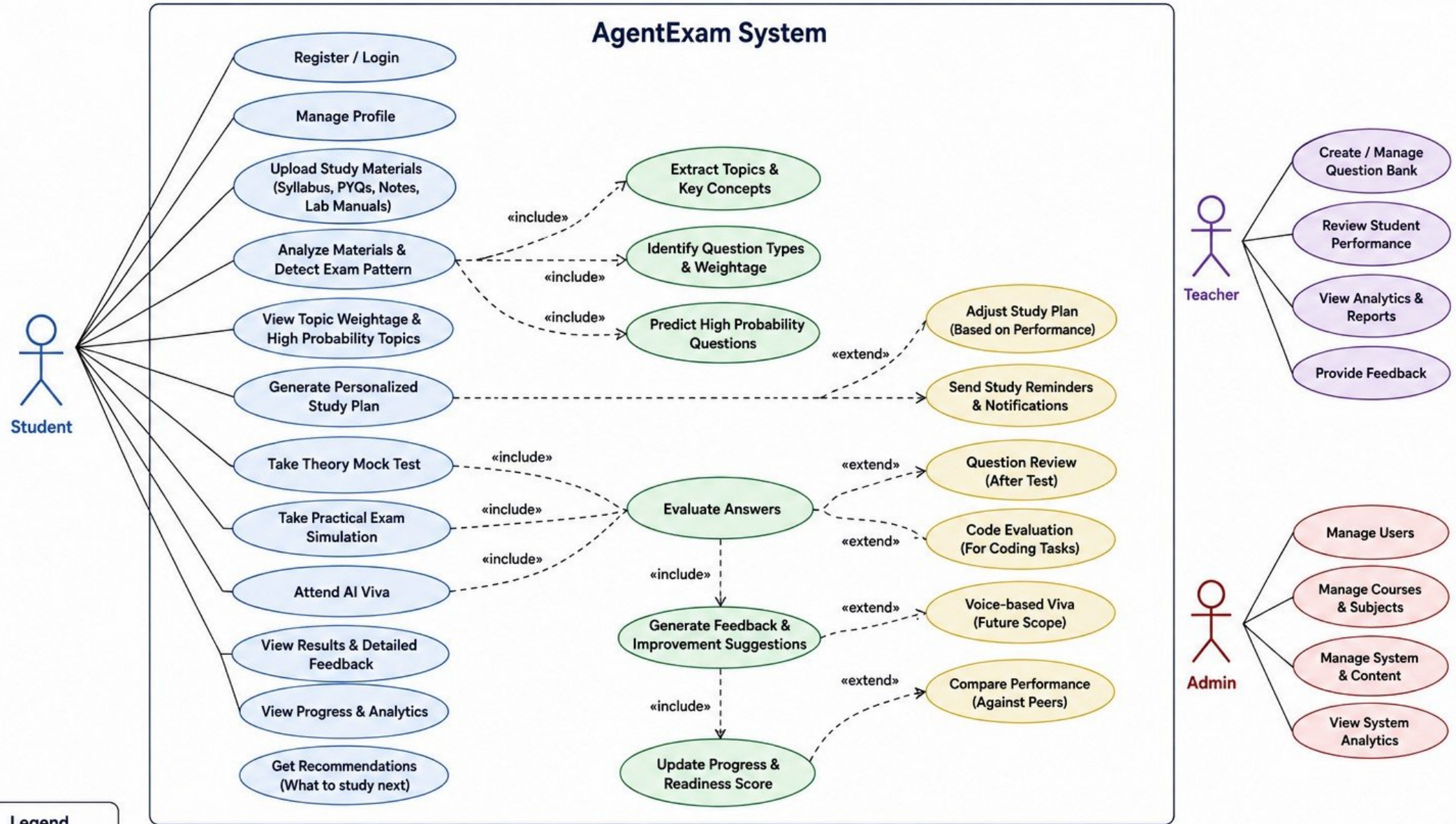


AGENTEXAM – SYSTEM ARCHITECTURE

AI-Powered Autonomous Exam Preparation and Assessment System (RAG Based)

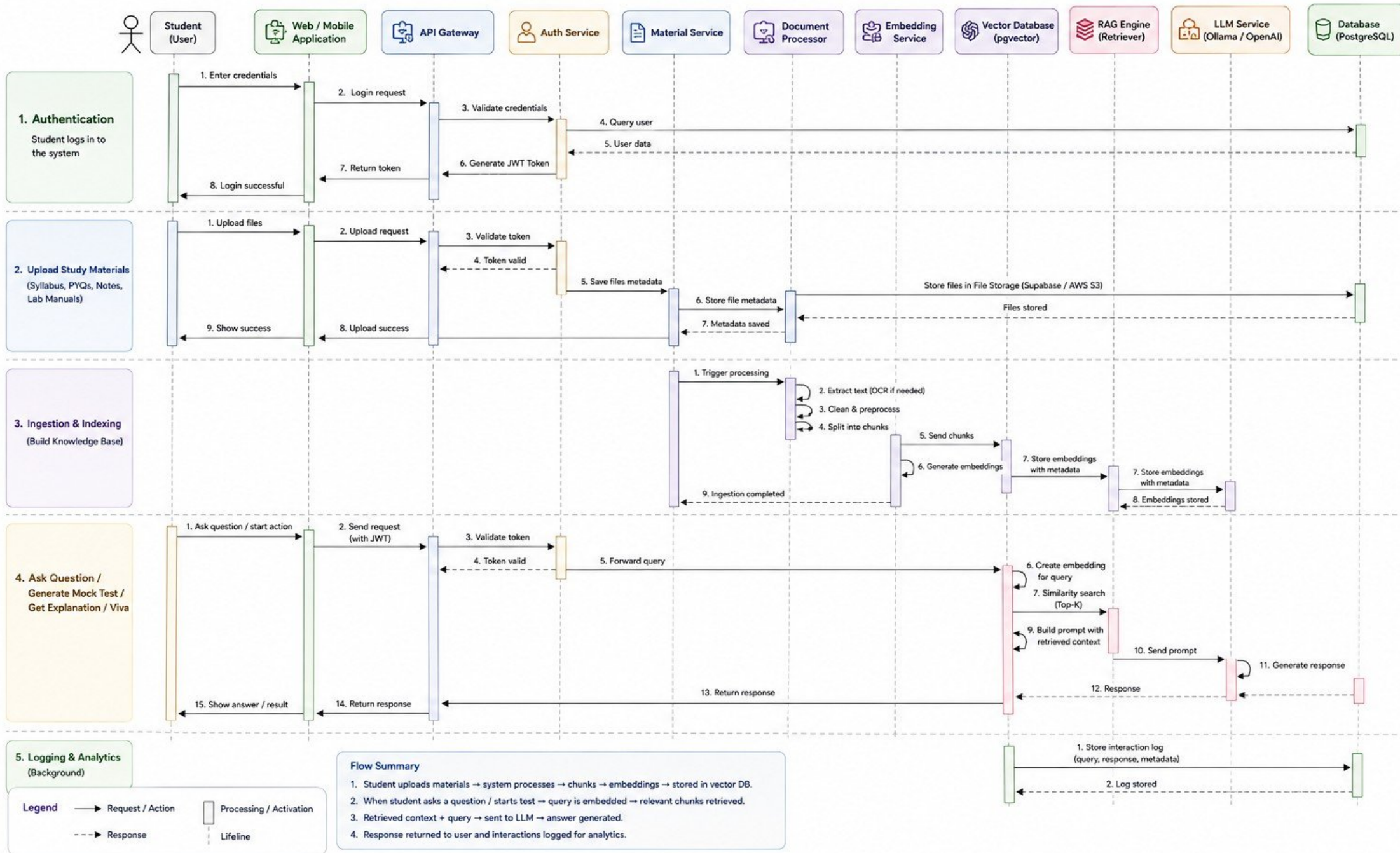


AgentExam – Use Case Diagram



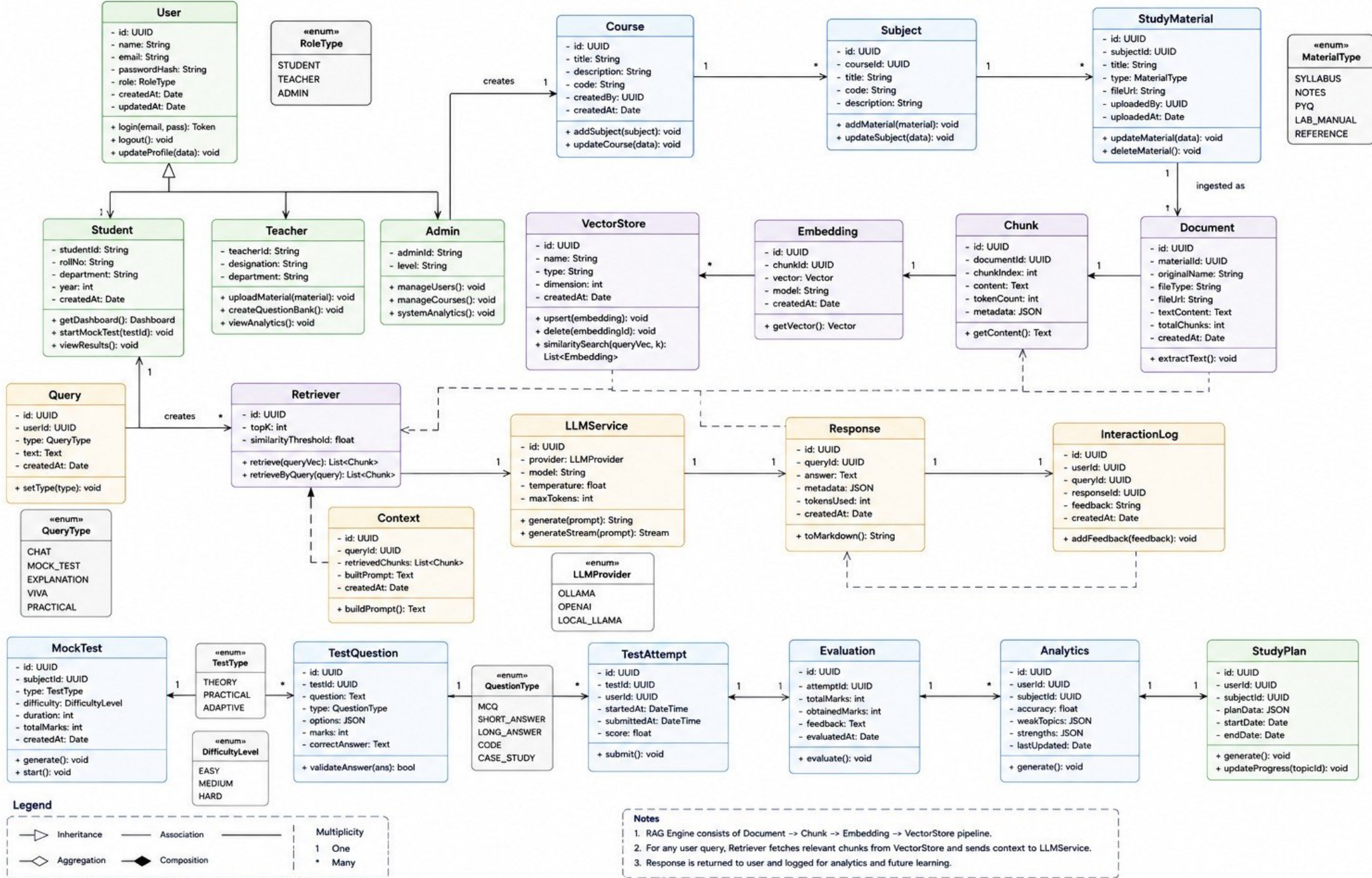
AgentExam – Sequence Diagram (RAG Based)

AI-Powered Autonomous Exam Preparation and Assessment System



AgentExam – Class Diagram (RAG Based)

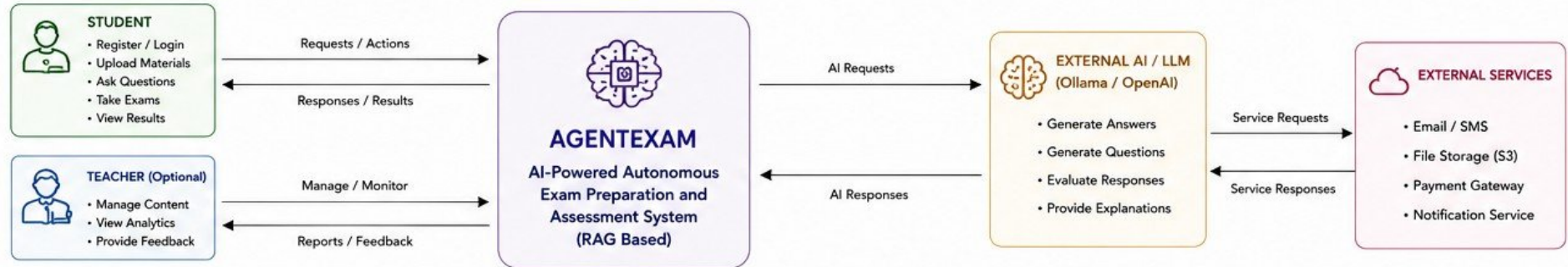
AI-Powered Autonomous Exam Preparation and Assessment System



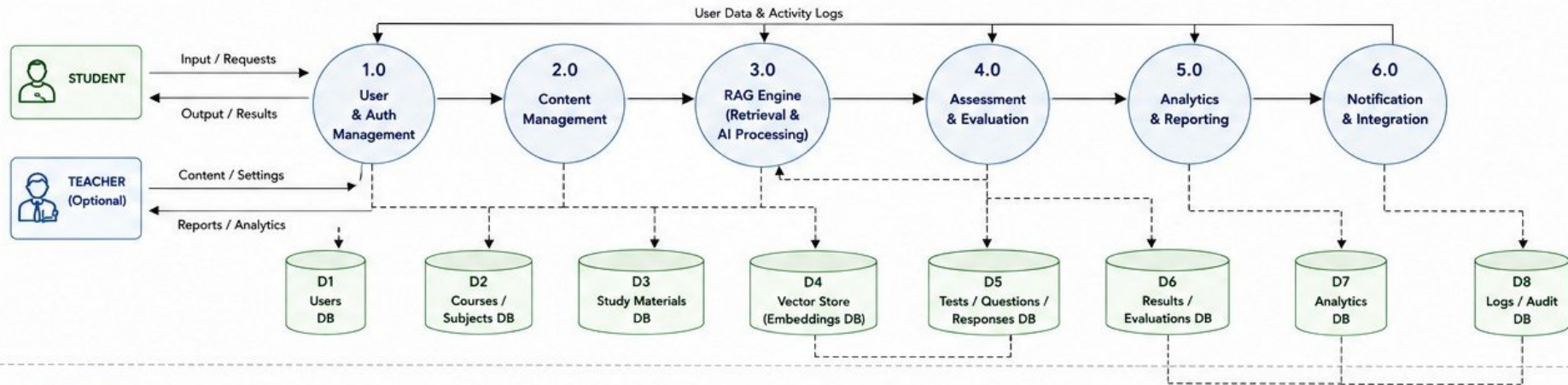
AGENTEXAM – DATA FLOW DIAGRAM (RAG BASED)

AI-Powered Autonomous Exam Preparation and Assessment System

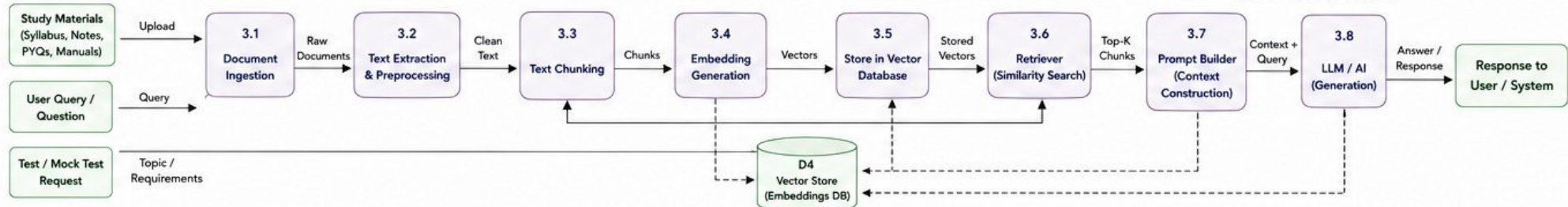
LEVEL 0 Context Diagram



LEVEL 1 Top Level DFD



LEVEL 2 Detailed DFD – RAG Engine (3.0)



LEGEND

External Entity

Main Process (Level 1)

Data Store

→ Data Flow

--- Data Flow (Read/Write)

→ Request / Input Flow

--- External Service Flow

Sub-Process (Level 2)

Data Store

DATA STORES DESCRIPTION

D1: User credentials, profiles, roles

D2: Courses, subjects, chapters, topics

D3: Uploaded materials, files, metadata

D4: Embeddings, vector representations

D5: Questions, tests, attempts, answers

D6: Evaluations, scores, feedback

D7: Analytics data, performance, trends

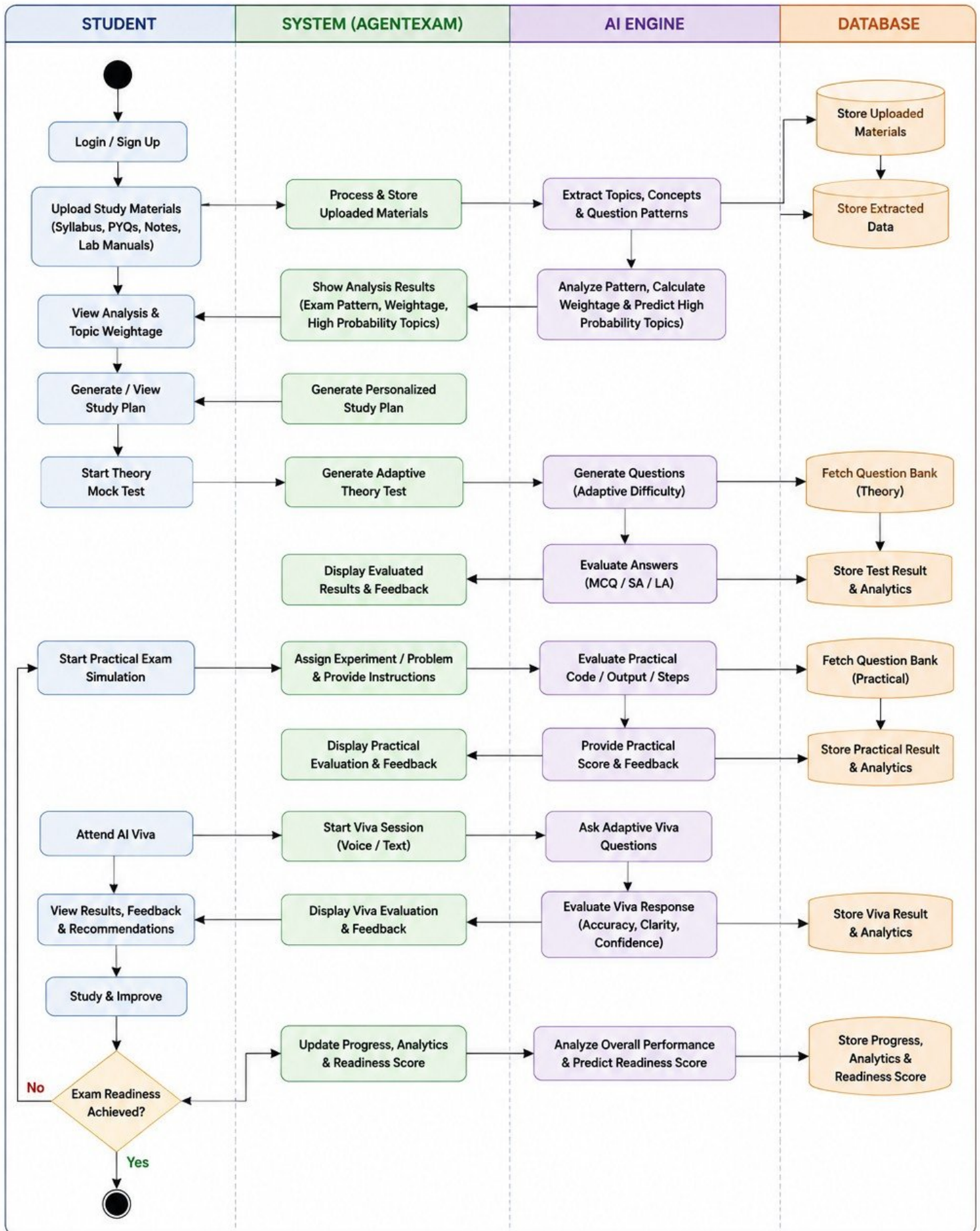
D8: System logs, activities, audit trails

NOTE

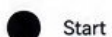
- Level 0 shows the system as a single process.
- Level 1 decomposes into major functional modules.
- Level 2 details the RAG Engine (core AI component).
- Dotted lines indicate data read/write to data stores.

AgentExam – Activity Diagram

(Student Exam Preparation & Assessment Flow)



LEGEND



Start



End



Student Activity



System Activity



AI Activity



Database



Decision

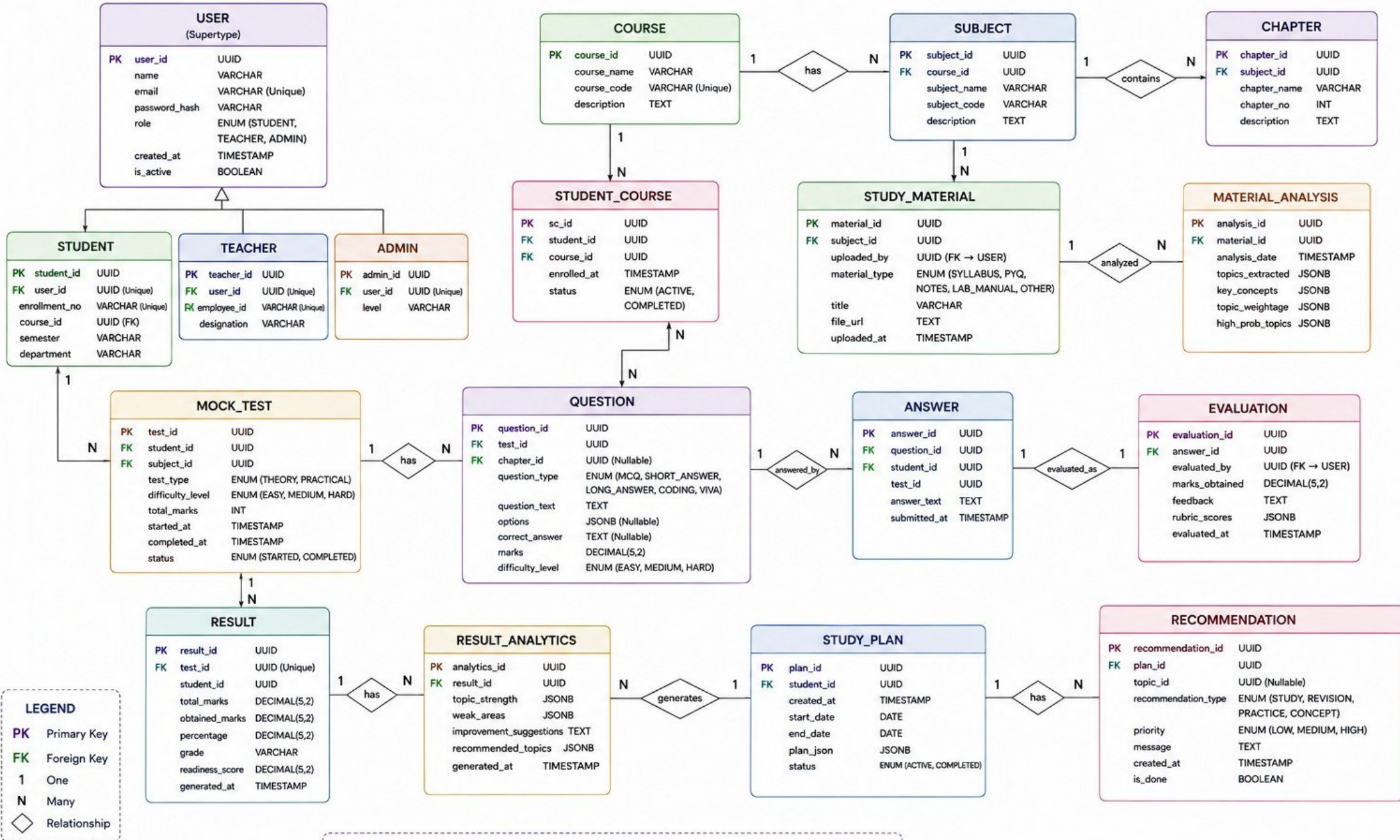


Control Flow

Notes:

The system continuously tracks performance and guides the student until they are exam ready.

AgentExam – ER Diagram



AGENTEXAM – MODULE DESIGN

AI-Powered Autonomous Exam Preparation and Assessment System (RAG Based)

